Problem Set 8

Problem 1. (Two Constraints)Find the extreme value of f subject to both constraints

(i) $f(x, y, z) = x + 2y$; $x + y + z = 1, y^2 + z^2 = 4$

(ii) $f(x, y, z) = 3x - y - 3z$; $x + y - z = 0$, $x^2 + 2z^2 = 1$

(iii) $f(x, y, z) = yz + xy$; $xy = 1$, $y^2 + z^2 = 1$

(iv) $f(x, y, z) = x^2 + y^2 + z^2$; $x - y = 1$, $y^2 - z^2 = 1$

Problem 2. (Global maxima and minima) Find the absolute maximum and minimum values of the following function in the specified region.

(i) $f(x, y) = 4xy^2 - x^2y^2 - xy^3$; R is the closed triangular region in the xy-plane with vertices $(0, 0)$, $(0, 6)$ and $(6, 0)$

(ii) $f(x, y) = e^{-x^2-y^2}(x^2 + 2y^2)$; $R = \{(x, y) : x^2 + y^2 \leq 1\}$

Problem 3. Evaluate the double integral by first identifying it as the volume of a solid.

(i) $\iint_R 3\, dA$, $R = \{(x, y) : 2 \leq x \leq 2, 1 \leq y \leq 6\}$

(ii) $\iint_R 5x\, dA$, $R = \{(x, y) : 0 \leq x \leq 5,\ 0 \leq y \leq 3\}$

Problem 4. Evaluate the following iterated integrals

(i) $\int_0^2 \int_0^4 y^2 e^{2x}\, dydx$

(ii) $\int_{-3}^3 \int_0^{\frac{\pi}{2}} (y + y^2 \cos x)\, dxdy$

(iii) $\int_1^3 \int_1^5 \ln y\, xy\, dydx$

(iv) $\int_0^1 \int_0^1 v(u + v^2)^4\, dudv$

(v) $\int_0^1 \int_0^1 xy\sqrt{x^2 + y^2}\, dydx$

(vi) $\int_0^2 \int_0^{\pi} r\sin^2\theta\, d\theta dr$

(vii) $\int_0^1 \int_0^1 \sqrt{s + t}\, dsdt$

(viii) $\int_1^2 \int_0^{2z} \int_0^{\ln x} xey\, dydxdz$

(*ix*) $\int_0^{\frac{\pi}{2}} \int_0^y \int_0^x \cos(x+y+z)\, dzdxdy$

(*x*) $\int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin y\, dydzdx$

Problem 5. Calculate the double integral of the following functions in the specified regions:

(*i*) $f(x,y) = \sin(x-y)\, dA$, $R = \{(x,y) : 0 \leq x \leq \frac{\pi}{2},\ 0 \leq y \leq \frac{\pi}{2}\}$

(*ii*) $f(x,y) = (y + xy^{-2})\, dA$, $R = \{(x,y) : 0 \leq x \leq 2,\ 1 \leq y \leq 2\}$

(*iii*) $f(x,y) = x^2 + 2y$, R is bounded by $y = 0$, $y = x^2$, $x = 1$

(*iv*) $f(x,y) = y^2$, R is the triangular region with vertices $(0,1), (1,2), (4,1)$.

(*v*) $f(x,y) = xy^2$, R is enclosed by $x = 0$ and $x = \sqrt{1-y^2}$

(*vi*) $f(x,y) = \sin(x^2+y^2)$ R is the region in the first quardant between the circle $x^2+y^2 = a^2$ and $x^2+y^2 = b^2$ with $0 < a < b$

(*vii*) $f(x,y) = e^{-x^2-y^2}$ R is the region bounded by the semicircle $x = \sqrt{4-y^2}$ and y-axis

(*viii*) $f(x,y) = \arctan\frac{y}{x}$, $R = \{(x,y) : 1 \leq x^2+y^2 \leq 4, 0 \leq y \leq x\}$

Problem 6. Solve the following

(*i*) Find the volume of the solid bounded by the circular paraboloid $z = 1 - x^2 - y^2$ and xy-plane

(*ii*) Find the volume of the solid under the cone $z = \sqrt{x^2+y^2}$ and above the disk $x^2+y^2 \leq 4$

(*iii*) Find the volume of the solid below the paraboloid $z = 18 - 2x^2 - 2y^2$ and above the xy-plane

(*iv*) Find the volume of the solid enclosed by the hyperboloid $-x^2 - y^2 + z^2 = 1$ and the plane $z = 2$

(*v*) Find the volume of the solid inside the sphere $x^2+y^2+z^2 = 16$ and outside the cylinder $x^2+y^2 = 4$

Problem 7. Evaluate the triple integral of the following functions in the specified solids.

(*i*) $f(x,y,z) = y$, $V = \{(x,y,z) : 0 \leq x \leq 3, 0 \leq y \leq x, x-y \leq z \leq x+y\}$

(ii) $f(x,y,z) = \frac{z}{x^2+y^2}$, $V = \{1 \leq y \leq 4,\ y \leq z \leq 4,\ 0 \leq x \leq z\}$

(iii) $f(x,y,z) = \sin y$, V is the solid below the plane $z = x$ and above the triangular region with vertices $(0,0,0), (\pi,0,0)$, and $(0,\pi,0)$

(iv) $f(x,y,z) = xy$, V is bounded by the parabolic cylinders $y = x^2$ and $x = y^2$ and the planes $z = 0$ and $z = x + y$

(v) $f(x,y,z) = x^2$, V is the solid tetrahedron with vertices $(0,0,0), (1,0,0), (0,1,0)$, and $(0,0,1)$

Problem 8. In the following assume that a double integral reduces to the following repeated one-dimensional integrations. For each case draw the underlying region, and change the order of integration and write down the underlying limits:

(i) $\int_0^1 [\int_0^y f(x,y)dx]dy$

(ii) $\int_0^2 [\int_{2y}^{y^2} f(x,y)dx]dy$

(iii) $\int_1^e [\int_0^{\ln x} f(x,y)dx]dy$

Problem 9. Write three other iterated integrals that are equal to the given iterated integral.

$\int_0^1 \int_y^1 \int_0^y f(x,y,z)\, dxdzdy$

$\int_0^1 \int_0^{1-x^2} \int_0^{1-x} f(x,y,z)\, dydzdx$

$\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x,y,z)\, dzdydx$